

human

Where are the thatched roof houses in Exeter?

host_agent

Calling tool send_message with args : {'target': 'planning_agent', 'task_description': 'User wants to find thatched roof houses in Exeter. Provide the general steps to solve this geospatial query.'}

planning_agent

The area is Exeter, so there will be 1 entry for Exeter. The entity to search for is thatched roof houses, and we need to find as many entries as possible for thatched roof houses in Exeter.

output steps: ["Find Exeter 1 area", "Search for thatched roof houses in Exeter as many search results"]

host_agent

Tool send_message returned result : The area is Exeter, so there will be 1 entry for Exeter. The entity to search for is thatched roof houses, and we need to find as many entries as possible for thatched roof houses in Exeter.

output steps: ["Find Exeter 1 area", "Search for thatched roof houses in Exeter as many search results"]

Calling tool send_message with args : {'target': 'named_area', 'task_description': 'Find the area polygon for Exeter.'}

named_area_agent

Calling tool call_os_ngd with args : {'filters': None, 'bbox': None, 'polygon_or_point': True, 'street_address': None, 'filename': 'exeter_area_polygon'}

Tool call_os_ngd returned result : ['Search results have been found. \n Artifact generated is : exeter_area_polygon,\n Description is : A geopandas dataframe containing named area data with no filters applied for the query (so remember no search is performed here so spurious entities present) Find the area polygon for Exeter..,\n count of records fetched : 98.\n ',]

Calling tool send_message with args : {'target': 'data_analysis_agent', 'task_description': "Filter the artifact 'exeter_area_polygon' to find the specific polygon for Exeter. Save the filtered results as 'filtered_exeter_area_polygon'."}

data_analysis_agent

Calling tool generate_metadata_for_artifacts with args : {'artifact_names': ['exeter_area_polygon']}

Tool Code Executor

```
import matplotlib
matplotlib.use('Agg')
def filter_exeter_area_polygon(data: list):
    gdf = data[0]
    # Filter for Exeter using multiple relevant columns
    filtered = gdf[((gdf['name1_text'].str.lower() == 'exeter') |
                    (gdf['name1_text'].str.lower().str.contains('exeter')) |
                    (gdf['description'].str.lower() == 'city') |
                    (gdf['description'].str.lower() == 'town')) &
                  (gdf['descriptiongroup'].str.lower() == 'settlement')]
    summary = f"Filtered to {len(filtered)} polygon(s) for Exeter using name and description columns."
    artifact_name = 'filtered_exeter_area_polygon'
    artifact_description = 'GeoDataFrame containing only the polygon(s) representing Exeter, filtered by name and settlement description.'
    return [summary, artifact_name, artifact_description, filtered]
```

Tool Code Executor

```
import matplotlib
matplotlib.use('Agg')
def filter_exeter_area_polygon(data: list):
    gdf = data[0]
    # Filter for Exeter using multiple relevant columns
    filtered = gdf[((gdf['name1_text'].str.lower() == 'exeter') |
                    (gdf['name1_text'].str.lower().str.contains('exeter')) |
                    (gdf['description'].str.lower() == 'city') |
                    (gdf['description'].str.lower() == 'town')) &
                  (gdf['descriptiongroup'].str.lower() == 'settlement')]
    summary = f"Filtered to {len(filtered)} polygon(s) for Exeter using name and description columns."
    artifact_name = 'filtered_exeter_area_polygon'
    artifact_description = 'GeoDataFrame containing only the polygon(s) representing Exeter, filtered by name and settlement description.'
    return [summary, artifact_name, artifact_description, filtered]

# Call the function with the provided data list
filter_exeter_area_polygon([exeter_area_polygon])
```

Tool Code Executor

```

import matplotlib
matplotlib.use('Agg')
def filter_exeter_area_polygon(data: list):
    gdf = data[0]
    # Filter for Exeter using multiple relevant columns
    filtered = gdf[((gdf['name1_text'].str.lower() == 'exeter') |
                    (gdf['name1_text'].str.lower().str.contains('exeter')) |
                    (gdf['description'].str.lower() == 'city') |
                    (gdf['description'].str.lower() == 'town')) &
                  (gdf['descriptiongroup'].str.lower() == 'settlement')]
    summary = f"Filtered to {len(filtered)} polygon(s) for Exeter using name and description columns."
    artifact_name = 'filtered_exeter_area_polygon'
    artifact_description = 'GeoDataFrame containing only the polygon(s) representing Exeter, filtered by name and settlement description.'
    return [summary, artifact_name, artifact_description, filtered]

```

Tool code_executor returned result : ['Filtered to 2 polygon(s) for Exeter using name and description columns.',]

Output from agent data_analysis_agent : Filtered to 2 polygon(s) for Exeter using name and description columns.

Artifact name: filtered_exeter_area_polygon

Description: GeoDataFrame containing only the polygon(s) representing Exeter, filtered by name and settlement description.

If you need to view or analyze these polygons further, let me know!

Tool send_message returned result : ["Filtered to 2 polygon(s) for Exeter using name and description columns.\n\nArtifact name: filtered_exeter_area_polygon\n\nDescription: GeoDataFrame containing only the polygon(s) representing Exeter, filtered by name and settlement description.\n\nIf you need to view or analyze these polygons further, let me know!Additionally some data artifacts have been generated with names ['filtered_exeter_area_polygon'] and \n descriptions ['GeoDataFrame containing only the polygon(s) representing Exeter, filtered by name and settlement description.']", []]

named_area_agent

I found the area polygon for Exeter and filtered it to ensure accuracy. The filtered artifact is named `filtered_exeter_area_polygon`, which contains the specific polygon(s) representing Exeter.

If you need further analysis or details, feel free to ask!

Tool send_message returned result : ["I found the area polygon for Exeter and filtered it to ensure accuracy. The filtered artifact is named `filtered_exeter_area_polygon`, which contains the specific polygon(s) representing Exeter.\n\nIf you need further analysis or details, feel free to

ask!Additionally some data artifacts have been generated with names ['filtered_exeter_area_polygon'] and \n descriptions ['GeoDataFrame containing only the polygon(s) representing Exeter, filtered by name and settlement description.']", []

host_agent

Calling tool generate_metadata_for_all_artifacts with args : {}

Tool generate_metadata_for_all_artifacts returned result : {'exeter_area_polygon': 'A geopandas dataframe containing named area data with no filters applied for the query (so remember no search is performed here so spurious entities present) Find the area polygon for Exeter..', 'filtered_exeter_area_polygon': 'GeoDataFrame containing only the polygon(s) representing Exeter, filtered by name and settlement description.'}

Calling tool send_message with args : {'target': 'buildings', 'task_description': "Search for thatched roof houses within the area defined by the artifact 'filtered_exeter_area_polygon'."}

buildings_agent

Calling tool call_os_ngd with args : {'filters': None, 'bbox': 'filtered_exeter_area_polygon', 'polygon_or_point': True, 'street_address': None, 'filename': 'all_buildings_exeter_area'}

Tool call_os_ngd returned result : ['Multiple search results have been found in multiple datasets. A summary of each is provided.\n Artifacts generated are : ['buildings_all_buildings_exeter_area', 'buildingline_all_buildings_exeter_area', 'buildingpart_all_buildings_exeter_area'],\n Descriptions are : ["A geopandas dataframe containing building data with filters and bbox applied for the query Search for thatched roof houses within the area defined by the artifact 'filtered_exeter_area_polygon'. using filters None.", "A geopandas dataframe containing building line data with filters and bbox applied for the query Search for thatched roof houses within the area defined by the artifact 'filtered_exeter_area_polygon'. using filters None.", "A geopandas dataframe containing building part data with filters and bbox applied for the query Search for thatched roof houses within the area defined by the artifact 'filtered_exeter_area_polygon'. using filters None."],\n counts of records fetched are : [72764, 72292, 89675].\n ', [, ,]]

Calling tool send_message with args : {'target': 'data_analysis_agent', 'task_description': "From the artifact 'buildings_all_buildings_exeter_area', filter for houses as defined by: buildinguse_addresscount_residential > 0 and buildinguse_addresscount_total = 1. Then, within these houses, identify those with thatched roofs (look for roof material or description indicating 'thatch'). Save the filtered results as 'thatched_roof_houses_exeter_area'."}

data_analysis_agent

Calling tool generate_metadata_for_artifacts with args : {'artifact_names': ['buildings_all_buildings_exeter_area']}

Tool Code Executor

```
def filter_thatched_roof_houses_exeter_area(data: list):
    import pandas as pd
    import geopandas as gpd
    df = data[0]
    # Step 1: Filter for houses: residential > 0 and total = 1
    houses = df[(df['buildinguse_addresscount_residential'] > 0) & (df['buildinguse_addresscount_total'] == 1)]
    # Step 2: Identify thatched roofs
    # Use roofmaterial_primarymaterial and description columns
    mask_roof = houses['roofmaterial_primarymaterial'].str.contains('thatch', case=False, na=False)
    mask_desc = houses['description'].str.contains('thatch', case=False, na=False)
    thatched = houses[mask_roof | mask_desc]
    summary = f"Filtered {len(thatched)} thatched roof houses from {len(df)} buildings in the Exeter area."
    artifact_name = 'thatched_roof_houses_exeter_area'
    artifact_description = 'GeoDataFrame of houses (residential > 0, total = 1) in Exeter area with thatched roofs, identified by roof material or description.'
    return [summary, artifact_name, artifact_description, thatched]
```

Tool code_executor returned result : ['Filtered 29 thatched roof houses from 72764 buildings in the Exeter area.',]

Output from agent data_analysis_agent : Filtered 29 thatched roof houses from 72,764 buildings in the Exeter area. Artifact name: thatched_roof_houses_exeter_area Description: GeoDataFrame of houses (residential > 0, total = 1) in Exeter area with thatched roofs, identified by roof material or description.

Tool send_message returned result : ["Filtered 29 thatched roof houses from 72,764 buildings in the Exeter area.\nArtifact name: thatched_roof_houses_exeter_area\nDescription: GeoDataFrame of houses (residential > 0, total = 1) in Exeter area with thatched roofs, identified by roof material or description.Additionally some data artifacts have been generated with names ['thatched_roof_houses_exeter_area'] and \n descriptions ['GeoDataFrame of houses (residential > 0, total = 1) in Exeter area with thatched roofs, identified by roof material or description.']", []]

buildings_agent

Search Results: - 29 thatched roof houses were found within the area defined by the artifact 'filtered_exeter_area_polygon'.

Definition used for "house": buildinguse_addresscount_residential > 0 and buildinguse_addresscount_total = 1, as per the special scenario rules. Thatched roofs were identified by roof material or description indicating "thatch".

Filtered artifact name: thatched_roof_houses_exeter_area Description: GeoDataFrame of houses (residential > 0, total = 1) in Exeter area with thatched roofs, identified by roof material or description.

If you need further analysis or details about these houses, let me know!

Tool send_message returned result : ['Search Results:\n- 29 thatched roof houses were found within the area defined by the artifact \nfiltered_exeter_area_polygon\n.\n\nDefinition used for "house": buildinguse_addresscount_residential > 0 and buildinguse_addresscount_total = 1, as per the special scenario rules. Thatched roofs were identified by roof material or description indicating "thatch".\n\nFiltered artifact name: thatched_roof_houses_exeter_area\nDescription: GeoDataFrame of houses (residential > 0, total = 1) in Exeter area with thatched roofs, identified by roof material or description.\n\nIf you need further analysis or details about these houses, let me know!Additionally some data artifacts have been generated with names [\n'thatched_roof_houses_exeter_area\n'] and \n descriptions [\n'GeoDataFrame of houses (residential > 0, total = 1) in Exeter area with thatched roofs, identified by roof material or description.\n'], []]

host_agent

Calling tool send_message with args : {'target': 'plotting_agent', 'task_description': "Plot the thatched roof houses in Exeter using the artifact 'thatched_roof_houses_exeter_area' and overlay the Exeter area boundary from 'filtered_exeter_area_polygon'."}

plotting_agent

Calling tool generate_metadata_for_artifacts with args : {'artifact_names': ['thatched_roof_houses_exeter_area', 'filtered_exeter_area_polygon']}

Tool Code Executor

```
import matplotlib
matplotlib.use('Agg')
def plot_thatched_roof_houses_exeter(data: list):
    import geopandas as gpd
    import folium

    # Unpack data
    thatched_gdf = data[0].copy()
    exeter_gdf = data[1].copy()

    # Ensure CRS is EPSG:4326 for folium
    if thatched_gdf.crs is None or thatched_gdf.crs.to_epsg() != 4326:
        thatched_gdf = thatched_gdf.to_crs(epsg=4326)
    if exeter_gdf.crs is None or exeter_gdf.crs.to_epsg() != 4326:
```



```

exeter_gdf = exeter_gdf.to_crs(epsg=4326)

# Filter Exeter city polygon (use name1_text == 'Exeter' only)
exeter_city = exeter_gdf[exeter_gdf['name1_text'] == 'Exeter']

# Spatial join: thatched houses within Exeter city boundary
thatched_in_exeter = gpd.sjoin(thatched_gdf, exeter_city, predicate='within', how='inner')

# Create folium map centered on Exeter
centroid = exeter_city.geometry.iloc[0].centroid
m = folium.Map(location=[centroid.y, centroid.x], zoom_start=13)

# Add Exeter boundary
folium.GeoJson(
    exeter_city.geometry.iloc[0],
    name='Exeter City Boundary',
    style_function=lambda x: {'color': 'blue', 'weight': 3, 'fillOpacity': 0.05}
).add_to(m)

# Add thatched roof houses
for _, row in thatched_in_exeter.iterrows():
    folium.GeoJson(
        row['geometry'],
        name='Thatched Roof House',
        style_function=lambda x: {'color': 'green', 'weight': 2, 'fillOpacity': 0.4}
    ).add_to(m)
    folium.Popup(f"Area: {row['geometry_area_m2']:.1f} m²").add_to(
        folium.GeoJson(row['geometry'])
    )

m.save('thatched_roof_houses_exeter_map.html')

# Prepare summary
summary_rows = thatched_in_exeter[['geometry_area_m2', 'roofmaterial_primarymaterial']].head(5)
summary = f"Plotted {len(thatched_in_exeter)} thatched roof houses within Exeter city boundary. First 5 entries:\n" + summary_rows.to_string(index=False)

return [
    summary,
    'thatched_roof_houses_exeter_map.html',
    'A folium map showing all thatched roof houses within the Exeter city boundary, overlaid with the Exeter area polygon.',
    'thatched_roof_houses_exeter_map.html'
]

```

Tool code_executor returned result : ['Plotted 29 thatched roof houses within Exeter city boundary. First 5 entries:\n geometry_area_m2\n roofmaterial_primarymaterial\n 86.108 Thatch\n 64.048 Thatch\n 140.678 Thatch\n 56.887 Thatch\n 154.689 Thatch',]

Output from agent plotting_agent : Plotted 29 thatched roof houses within Exeter city boundary. First 5 entries: geometry_area_m2
roofmaterial_primarymaterial 86.108 Thatch 64.048 Thatch 140.678 Thatch 56.887 Thatch 154.689 Thatch

An interactive folium map has been generated showing all thatched roof houses within the Exeter city boundary, overlaid with the Exeter area polygon.

Artifact name: thatched_roof_houses_exeter_map.html

Artifact description: A folium map showing all thatched roof houses within the Exeter city boundary, overlaid with the Exeter area polygon.

Artifact data: thatched_roof_houses_exeter_map.html

Tool send_message returned result : ["Plotted 29 thatched roof houses within Exeter city boundary. First 5 entries:\n geometry_area_m2\n roofmaterial_primarymaterial\n 86.108 Thatch\n 64.048 Thatch\n 140.678 Thatch\n 56.887 Thatch\n 154.689 Thatch\n\nAn interactive folium map has been generated showing all thatched roof houses within the Exeter city boundary, overlaid with the Exeter area polygon.\n\nArtifact name: thatched_roof_houses_exeter_map.html \nArtifact description: A folium map showing all thatched roof houses within the Exeter city boundary, overlaid with the Exeter area polygon. \nArtifact data: thatched_roof_houses_exeter_map.html\nAdditionally some data artifacts have been generated with names ['thatched_roof_houses_exeter_map.html'] and \n descriptions ['A folium map showing all thatched roof houses within the Exeter city boundary, overlaid with the Exeter area polygon.'], []

host_agent

There are 29 thatched roof houses identified within the Exeter city boundary. These houses were found by searching for residential buildings with thatched roofs in the official buildings database for Exeter.

- The search used the precise Exeter area boundary.
- Thatched roofs were identified by their roof material or description.

A map has been generated showing the locations of all thatched roof houses within Exeter, overlaid with the city boundary. If you would like to view or download this map, let me know! If you need more details about any specific house or area, I can provide that as well.